

# 45 Days Summer Training Program

## Topic - Basic Python Assignment - 1

Q.1. Explain the difference between the following data type with examples:

- Integer (int)  
used to store whole numbers (positive or negative)  
Ex: age = 17  
marks = -5  
print(type(age))  
o/p: <class 'int'>
- Float (float)  
used to store decimal no..  
Ex: price = 39.5  
print(type(price))  
o/p: <class 'float'>
- String (str)  
used to store text or characters.  
Ex: name = "sakshi"  
print(type(name))  
o/p: <class 'str'>
- Boolean (bool)  
used to store true or false values.  
Ex: is-student = True  
print(type(is-student))  
o/p: <class 'bool'>

Q.2. Write a python program to create three variables:

- name
- age
- city

Store your details in them and print the values.

Sample output: Name: Gargi  
Age: 20  
City: Jaipur

```
Name = "Gargi"  
Age = 20  
City = "Jaipur"  
print("Name: ", Name)  
print("Age: ", Age)  
print("City: ", City)
```

Q.3. Write a python program that:

Takes a user's name as input.

Print the name in uppercase.

Prints the total number of characters in the name.

```
name = "Sakshi"
Print("Enter your Name: ", name)
Print("Name in uppercase: ", name.upper())
Print("Total number of characters: ", len(name))
```

OP: Enter your Name: Sakshi  
Name in uppercase: SAKSHI  
Total number of characters: 6

Q.4. Explain any five commonly used string methods in Python with examples.

① String Comparison  
Comparing two strings to check whether they are equal, different or which one comes first alphabetically.

```
Ex: s1 = "abcd"
    s2 = "Sa"
    if s1 > s2:
        print("s1 > s2")
    else:
        print("s2 > s1")
```

② String Slicing:

In the string slicing we extract a substring from a larger string.

```
ss String[start:end:step]
String = "Hello, world!"
Substring = String[-6:]
print("Substring")
```

### ③ String Splitting

we can split a string into a list of substring using `split()` method.

```
str = "The quick brown fox, jumps over the lazy  
dog"
```

```
words = str.split(",")
```

```
print(words)
```

```
words = str.split()
```

```
string = "Quick, Brown, Fox"
```

```
xyz = string.split(",")
```

### ④ String Joining

we can join a list of strings in a single string by using `join` method.

```
Ex: str = ['Quick', 'Brown', 'Fox']
```

```
string = ','.join(str)
```

```
print(string)
```

### ⑤ String Stripping

we can remove white spaces characters from the beginning and/or end of a string using `strip` method.

```
Ex: str = " Hello, world! "
```

```
stripstr = str.strip() print(str)
```

Q.5. Create a list containing the name of five fruits

- Print the complete list.
- Print the first & last element
- Print the total number of items in the list

```
Fruits = ["Apple", "Mango", "Banana", "Orange", "Grapes"]
```

```
print("Complete list:", Fruits)
```

```
print("First Element and the last element:",
```

```
    Fruits[0], Fruits[-1])
```

```
print("Total number of items:", len(Fruits))
```

O/p : Complete list: ['Apple', 'Mango', 'Banana', 'Orange', 'Grapes']

First Element and the last Element: Apple  
Grapes

Total number of Items: 5

Q.6. Write a python program to:

- Create a list of numbers [10, 20, 30, 40, 50]
- Add 60 to the list
- Remove 20 from the list
- Print the updated list.

```
lst = [10, 20, 30, 40, 50]
```

```
print(lst)
```

```
lst.append(60)
```

```
print(lst)
```

O/p : [10, 20, 30, 40, 50]

[10, 20, 30, 40, 50, 60]

[10, 30, 40, 50, 60]

```
lst.remove(20)
```

```
print(lst)
```

Q.7. What is Artificial intelligence (AI)? Explain its importance and mention any four real-life application of AI.

AI is a branch of data science it enables the machines and computers to perform tasks

that normally require human intelligence.

Importance of AI is:

- Improves accuracy and reduces human error.
- Works fast from the human being.
- Increase productivity.
- Better in decision making.
- Processes large amount of data quickly.

Real-life example:

① Virtual Assistants: Google assistant, Siri.

② Healthcare

③ Self-Driving vehicles

④ Automatic Smart traffic light

Q.8. Identify whether the following are examples of AI and explain why.

• Chatgpt:

Yes its a AI tools. why, because it understand the language, answer questions and generates human like text using machine learning.

• Google Maps Route Prediction:

Yes this is also AI. Because it analyses the traffic, road conditions, calculate the time and past data to predict the best route.

• Calculator:

no, not an AI tools because it only calculates based on some fixed rules it does not make decisions.

- Netflix recommendations:  
Yes, A+. Because it recommends you video, dramas on the basis of your past data or history.

- Voice Assistants (Alexa/Siri):  
Yes, Because they understand our language then answer our question and perform tasks.